

LinguaTriage: Cross-Lingual Transfer and African Language Pretraining for Low-Resource Medical Triage in Lingala

Intra-Family Bantu Pretraining for 45M Speakers

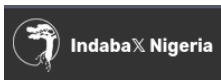
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Indaba X Nigeria 2026



Presentation Roadmap

Problem Statement

Methodology

Results

Conclusion & Impact

The Challenge: Zero NLP Resources for 45M Speakers

The Healthcare Crisis

- ▶ **DRC:** 0.09 physicians per 1,000 people
- ▶ One of the **lowest ratios globally** (WHO, 2022)
- ▶ Automated triage could save lives

The Language Gap

- ▶ **Lingala:** 45+ million speakers
- ▶ **Zero** annotated clinical corpus
- ▶ **Zero** supervised classification benchmark
- ▶ **Zero** automated triage system

Our Research Question

Can we build a clinically safe triage system for a language with **zero NLP resources**?

The Dialectal Challenge (Unique to Lingala)

Lingala is NOT monolithic

1. **Kinshasa Lingala:** Urban, French borrowing, aggressive contraction
2. **Equator region:** Classical variety, -djali paradigm
3. **Kisangani region:** Eastern variety, Swahili contact

Example: "I have a strong fever"

Nazali na fiere makasi (Kinshasa)

Naza na fiere makasi (Colloquial)

Naali na fiere makasi (Interior)

Nadjali na fiere makasi (Equator)

Same meaning → Tokenizer sees 4 different inputs!

Methodology Overview: Data to Model Pipeline

Step 1

Data Collection
616 samples
3 urgency levels

Augmentation
Stopword-aware
41→236
Emergency

Evaluation
Safety Metrics
Emergency
Recall

Model Training
XLM-RFT,
XLM-RCL,
AfriBERTaFT

Three Model Architectures Compared

XLM-RFT

XLM-RCL

AfriBERTaFT

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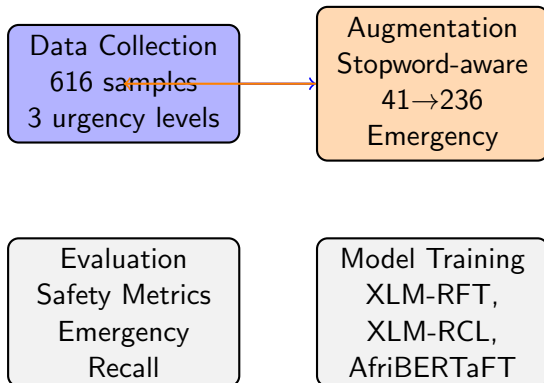
XLM-RFT

XLM-RCL

AfriBERTaFT

Methodology Overview: Data to Model Pipeline

Step 2: Augmentation



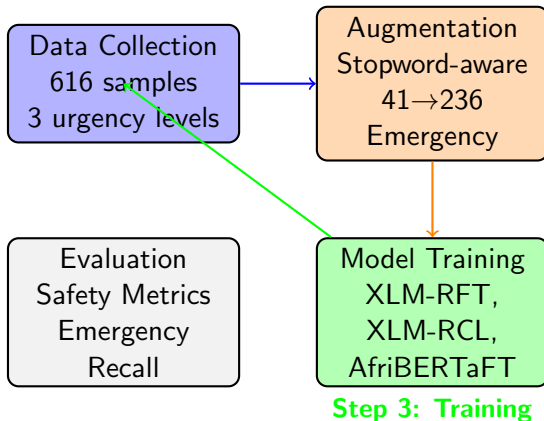
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XLM-RFT

XLM-RoBERTa fine-tuned

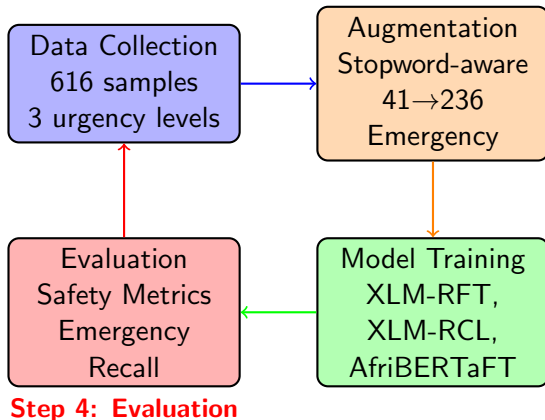
XLM-RCL

French → Lingala transfer

AfriBERTaFT

Bantu-family pretraining

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Three Model Architectures Compared

XLM-RFT

XLM-RoBERTa fine-tuned
125M parameters

XLM-RCL

French → Lingala transfer
Two-stage

AfriBERTaFT

Bantu-family pretraining
126M parameters **(Winner!)**

AfriBERTaFT Selected as Best Model

Dataset: From Severe Imbalance to Balanced

Pre-Augmentation		
Class	Count	%
Low	450	73.1%
Moderate	107	17.4%
Emergency	59	9.6%

Post-Augmentation		
Class	Train	Test
Emergency	236	10
Moderate	74	17
Low	315	68

Augmentation Pipeline (3 Strategies)

1. **Stopword-aware synonym substitution** (30-word medical lexicon)
2. **Clause reordering** (syntactic diversity)
3. **Urgency-prefix injection**

Emergency training samples: 41 → 236 (5.8x increase)

Novel Resource: First Lingala Medical Stopword List

Category	Count	Examples
Subject pronouns	8	na, a, ba
Copula & auxiliaries	47	nazali, naza, naali, nadjali
Negation	3	te, ata te
Adverbs	21	mingi, moke, makasi

Two Derived Sets

STOPWORDS_FULL:

130 tokens

STOPWORDS_MEDICAL:

Retains severity adverbs
(makasi, mingi, moke,
etc.)

First Lingala stopwords resource in any NLP library. Compiled from grammar references and validated by native speakers.

Main Results: Test Set Performance

Model	Acc.	F1	Recall	Err
SVM Baseline	0.926	0.834	1.000	0
XLNet-RFT	0.905	0.825	0.800	0
XLNet-RCL	0.853	0.669	0.900	0.100
AfriBERTaFT	0.990	0.974	1.000	0

Model Comparison

Performance ↑

Key Finding

AfriBERTaFT: Perfect Emergency Recall + Zero Critical Errors

XLNet-RCL: Only model with critical error (Emergency → Low)

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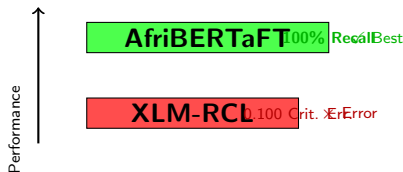
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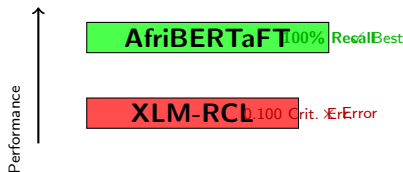
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Model Comparison



Winner: AfriBERTaFT

Key Finding

AfriBERTaFT: Perfect Emergency Recall + Zero Critical Errors

Confusion Matrix Comparison

SVM Baseline				
	E	M	L	
E	10	0	0	
M	2	11	4	
L	0	0	68	

XLM-RFT				
	E	M	L	
E	8	2	0	
M	2	11	4	
L	0	1	67	

AfriBERTaFT				
	E	M	L	
E	10	0	0	
M	0	15	2	
L	0	0	68	

XLM-RCL (not shown): 1 Emergency → Low (**CRITICAL ERROR**)

The Domain Shift Crisis & Solution

External Evaluation (Forum Data)

20-29% Accuracy
(essentially random)

Models predicted "Emergency" for nearly all inputs.

The Cure: +100 In-Domain Examples

79% Accuracy
Macro-F1: 0.726

Mixed 100 forum examples with training data.

Critical Finding

**Minimal target-domain exposure ;
Architectural choices**

- ▶ Before: 29% accuracy (no forum data)
- ▶ After: 79% accuracy (+100 forum examples)

Impact: What This Means for African AI

1. **First-ever Lingala NLP benchmark**

- ▶ 616 annotated samples, 3 urgency levels
- ▶ First Lingala stopwords resource (130 tokens)

2. **Typological proximity & Scale**

- ▶ AfriBERTa (11 African languages) outperformed XLM-R (100 languages)
- ▶ Lesson: Prioritize language-family-aware models

3. **Domain adaptation is critical**

- ▶ 100 in-domain examples = 29% → 79% accuracy
- ▶ Architectural gains cannot fix domain mismatch

Future Directions

Immediate Next Steps

- ▶ Expand dataset with natural clinical text
- ▶ LoRA fine-tuning for low-resource deployment
- ▶ Prospective evaluation with health workers in DRC

Long-term Vision

- ▶ Extend to other Central African Bantu languages
- ▶ Multimodal triage (voice + text)
- ▶ Integration with DRC health information systems

Code, Data, and Stopwords:

github.com/patrick-tenga/lingua-triage

Thank You

Thank You!

Questions?

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